Underexplored Mechanisms in Muon

Direction correction, learning-rate robustness, and momentum timing

Nevo Jonathan Sznajder and Itay Asolin | Mentors: Tehila Dahan and Kfir Y. Levy

This project studies three modular changes to the matrix-parameter update path of **Muon**: a MARS-inspired gradient-difference correction, time-varying momentum, and a history estimator inspired by Mu²SGD. The goal is not to claim a universally better optimizer, but to isolate which mechanisms matter, under which learning-rate policies, and whether their effects transfer across architectures and datasets.

Controlled experimental setting

- **ResNet-18 / CIFAR-100:** broad, inexpensive ablations.
- **TinyViT-5M / ImageNet-100:** cross-architecture confirmation.
- 50 epochs, batch size 128, evaluation every 5 epochs.
- Central comparisons use seeds 42, 43, and 44; error bars are sample SD.
- Rank-2-or-higher tensors use Muon. Biases, normalization parameters, and other rank-below-2 tensors use the same auxiliary AdamW branch in every Muon-family run.

Optimizer and regularization settings

Muon matrix weight decay: $\mathbf{5\times10^{-4}}$. Global gradient clipping: **1.0**. Auxiliary AdamW: LR $\mathbf{10^{-3}}$, betas **(0.9, 0.999)**, epsilon $\mathbf{10^{-8}}$, and no weight decay.

MARS-inspired correction: $\mathbf{c_t = g_t + gamma\ (g_t - g_{t-1})}$. Momentum schedules change **beta_t** over time before Muon's approximate polar map. The Mu²-inspired variant is explicitly not a full reproduction of Mu²SGD.

Main findings

MARS gives **small improvements in selected higher-LR conditions**. On the clean ResNet grid, the best tested gain was **+0.46 ± 0.22 percentage points** at matrix LR 0.04 and gamma=0.25. A focused TinyViT comparison gave **+0.34 ± 0.36 pp**.

Late momentum decay is the strongest result. For TinyViT at matrix LR 0.0025, decaying beta_t from 0.98 to 0.50 over epochs 36-48 produced a matched gain of +1.58 ± 0.33 pp under constant LR, with an endpoint of 85.36 ± 0.07%.

The same intervention under cosine LR changed the endpoint by **-0.31 ± 0.31 pp**. The LR policy therefore controls whether the refreshed direction still has enough step budget to change the final solution.

MARS and late momentum decay can coexist, but the remaining MARS increment after decay was only **+0.18 ± 0.29 pp**; the mechanisms are not reliably additive.

The Mu²-inspired correction changes pre- and post-polar directions, yet provides **no consistent accuracy improvement**. A visible geometric change is not sufficient evidence of a better descent direction.

Interpretation and scope

Muon largely normalizes singular-value scale, so the geometry and memory of the direction supplied to it remain important. Lowering **beta_t** shortens effective directional memory. Under a constant LR, the refreshed direction still produces meaningful movement; late in cosine decay, the remaining movement budget is much smaller.

The study covers two vision workloads, 50-epoch training, and three seeds in the central confirmations. Seed 42 also participated in exploration. The results support a focused optimizer-design note, not a universal superiority claim.

Public repository: github.com/nevogithub/muon-mechanisms-project